2SK-0970

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Assignment 2

1. i) $f(-2)=f(2)=\left\lfloor\frac{4}{3}\right\rfloor=1 \qquad f(3)=\left\lfloor\frac{9}{3}\right\rfloor=3$

$f(-1)=f(1)=\left\lfloor\frac{1}{3}\right\rfloor=0$

$f(0)=0 \qquad f(S)=\{0,1,3\}$

ii) $f(4)=\left\lfloor\frac{16}{3}\right\rfloor=5 \qquad f(5)=\left\lfloor\frac{25}{3}\right\rfloor=8$

$f(S)=\{0,1,3,5,8\}$

iii) $f(7)=\left\lfloor\frac{49}{3}\right\rfloor=16 \qquad f(11)=\left\lfloor\frac{121}{3}\right\rfloor=40$

$f(S)=\{0,8,16,40\}$

iv) $f(10)=\left\lfloor\frac{100}{3}\right\rfloor=33 \qquad f(14)=\left\lfloor\frac{196}{3}\right\rfloor=65$

$f(6)=\left\lfloor\frac{36}{3}\right\rfloor=12 \qquad f(S)=\{1,12,33,65\}$

2. $\lfloor -x \rfloor = -\lceil x \rceil$

$n-1 \le x \le n$

2. Multiply the inequality by -1

$-n \le -x \le -n+1$

$\lfloor -x \rfloor = -n$

$\lceil x \rceil = n$

$\therefore -\lceil x \rceil = -n$

$-n$ for both sides

so LHS = RHS

Teacher's Signature

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$\lceil -x \rceil = -\lfloor x \rfloor$

$n \leq x \leq n+1$ $\quad \lfloor x \rfloor = n \Rightarrow -\lfloor x \rfloor = -n$

$\therefore -(n+1) \leq -x \leq -n$ $\quad \lceil -x \rceil = -n$

LHS = RHS as $-n = -n$

3. a) $f \circ g$

$2(3a+2)+3$

$6a+4+3$

$6a+7$

$g \circ f$

$3(2a+3)+2$

$6a+9+2$

$6a+11$

b) f: injective as if $2x+3 = 2y+3$, then $x = y$

Not surjective as outputs are only odd numbers

Hence, it is not invertible as it is not surjective meaning a complete inverse function cannot be mapped back onto all integers

g: injective as if $3x+2 = 3y+2$, then $x = y$

Not surjective as only outputs integers that leave a remainder of 2 when divided by 3.

Hence, it is not invertible as it is not a bijective.

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4. a) $\{(0,0),(1,1),(2,2),(3,3)\}$

b) $\{(1,3),(2,2),(3,1),(4,0)\}$

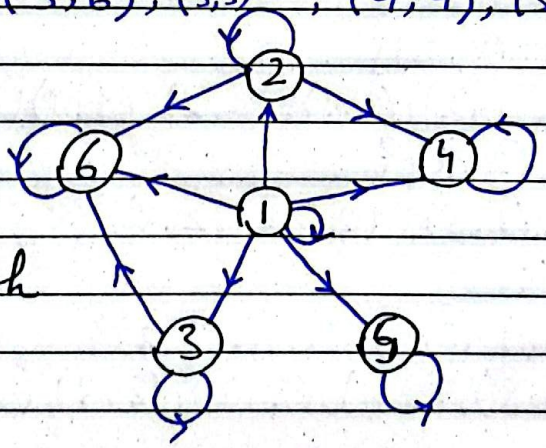
c) $\{(1,0),(2,0),(2,1),(3,0),(3,1),(3,2),(4,0),(4,1),(4,2),(4,3)\}$

d) $\{(0,0),(1,0),(1,1),(1,2),(1,3),(2,0),(2,2),(3,0),(3,3),(4,0)\}$

e) $\{(0,1),(1,0),(1,2),(1,3),(2,1),(1,1),(2,3),(3,1),(3,2),(4,1),(4,3)\}$

f) $\{(1,2),(2,1),(2,2)\}$

5. $\{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,2),(2,4),(2,6),(3,6),(3,3),(4,4),(5,5),(6,6)\}$



Digraph

1	1	1	1	1	1
1	1	0	1	0	1
1	0	1	0	0	1
1	1	0	1	0	0
1	0	0	0	1	0
1	1	1	0	0	1

Matrix

R is reflexive, antisymetric and transitive.
∴ it is Partial Order
& not equivalence

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6. part	reflexive	symetric	anti	transitive
a.	No	No	No	Yes
b.	Yes	Yes	No	Yes
c.	No	Yes	No	No
d.	No	No	Yes	No
e.	Yes	Yes	Yes	Yes
f.	No	No	No	No

7. part	reflexive	sym.	anti.	Asy.	irref	trans.
a	No	No	Yes	Yes	Yes	Yes
b	Yes	Yes	No	No	No	Yes
c	Yes	Yes	No	No	No	Yes
d	Yes	Yes	No	No	No	Yes

8. a) $a=b$ $R=\{(a,b) \mid a=b\}$ b) $\{(1,2),(2,1),(3,1)\}$
c) Universal Relation d) $R=\{(a,b) \mid a=b+1\}$

9. $R_1=\{(2,1),(3,1),(3,2)\}$
$R_2=\{(1,1),(2,1),(2,2),(3,1),(3,2),(3,3)\}$
$R_3=\{(1,2),(1,3),(2,3)\}$
$R_4=\{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}$
$R_5=\{(1,1),(2,2),(3,3)\}$
$R_6=\{(1,2),(1,3),(2,1),(2,3),(3,1),(3,2)\}$

a) $\{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)\}$

b) $\{(1,2),(1,3),(2,1),(2,3),(3,1),(3,2)\}$

c) $\{(1,2),(1,3),(2,3)\}$

d) $\{(1,2),(1,3),(2,3)\}$

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e) $\emptyset$

f) $\{(2,1),(3,1),(3,2)\}$

g) $\{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}$

h) $\{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}$

i) $\{(2,1),(3,1),(3,2)\}$

j) Universal set

10. i) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ ii) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ iii) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ iv) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

11. i) $\{(1,1),(1,3),(2,2),(3,1),(3,3)\}$

ii) $\{(1,2),(2,2),(3,2)\}$

iii) $\{(1,1),(1,2),(1,3),(2,1),(2,3),(3,1),(3,2),(3,3)\}$

12. • reflexive as $l(a)=l(a)$
• symetric as $l(a)=l(b)$ & $l(b)=l(a)$
• transective as if $l(a)=l(b)$ & $l(b)=l(c)$, then $l(a)=l(c)$

Equivalence Relation

13. 1. $\{(Ali, DB),(Bilal, AI),(Bilal, OS),(Ali, AI),$
$(Cathy, DB),(Cathy, OS),(David, DS),(David, OS)\}$

2. Domain = S Range = C

3. R is not a function, every student is mapped to more than one course.

4. $\{(DB, Ali),(AI, Ali),(AI, Bilal),(OS, Bilal),$
$(DB, Cathy),(DS, Cathy),(DS, David),(OS, David)\}$

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14. $f(0)=32$ $F(25)=77$ $f(100)=212$

Yes, F is one-to-one

$y=\frac{9}{5}C+32 \Rightarrow y-32=\frac{9}{5}C$

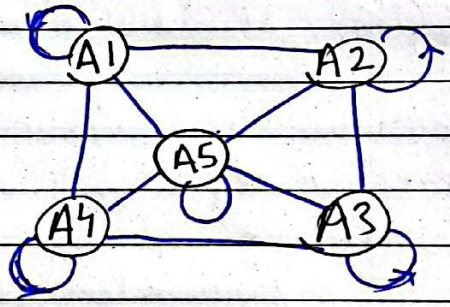
$\frac{5}{9}(y-32)=C \quad F^{-1}(f)=\frac{5}{9}(F-32)$

$F^{-1}(f)=\frac{5}{9}(95-32)=\frac{5}{9}(63)=35$

15. • reflexivity : $a \equiv a \pmod m$
• symetry : $a \equiv b \pmod m \quad b \equiv a \pmod m$
• transitivity : $a \equiv b \pmod m$ & $b \equiv c \pmod m \therefore a \equiv c \pmod m$

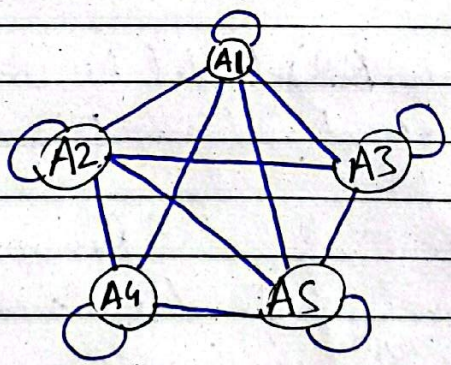
16. a)

1	1	0	1	1
1	1	1	0	1
0	1	1	1	1
1	0	1	1	1
1	1	1	1	1



b)

1	1	1	1	1
1	1	1	1	1
1	1	1	0	1
1	1	0	1	1
1	1	1	1	1



17. i) Vertices = 5

Ver	deg.
a	6
b	6
c	6
d	5
e	3

ii) Vertices:

Ver.
a
b
c
d
e
f
g
h
i

18. i) Vertices

Ver
a
b
c
d
e

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17. i) Vertices = 5 Edges: 13

Ver	deg.	Neighbourhood
a	6	{a, b, e}
b	6	{a, b, d, e}
c	6	{b, c, d}
d	5	{b, c, e}
e	3	{a, b, d}

ii) Vertices: 9 Edges: 12

Ver.	deg.	Neighbourhood
a	3	{c, e, i}
b	2	{e, h}
c	4	{a, e, g, i}
d	0	∅
e	6	{a, b, c, g}
f	0	∅
g	4	{c, e}
h	2	{b, i}
i	3	{a, c, h}

18. i) Vertices: 5 Edges: 15

Ver	In	Out
a	3	5
b	3	4
c	5	3
d	4	3
e	0	0

ii) Vertices: 4 Edges: 8

Ver	In	Out
a	1	3
b	3	4
c	3	0
d	1	1

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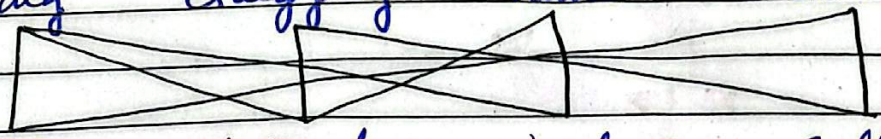
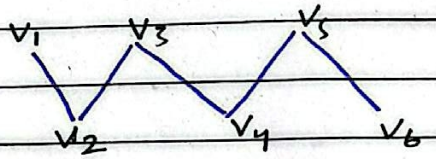
19. Ping Quiggley Russ Sitea



Hardware Networking Wireless Software

20. a) Not bipartite

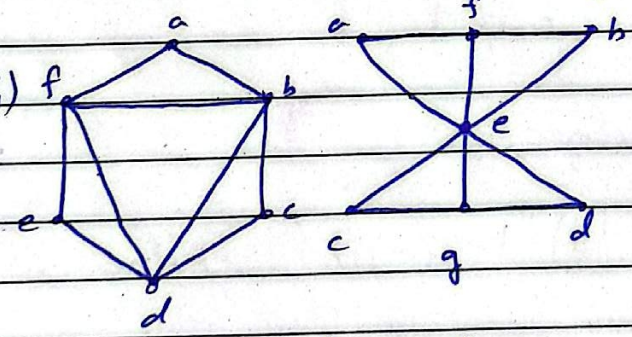
b) 

$\{v_1, v_3, v_5\}$
$\{v_2, v_4, v_6\}$

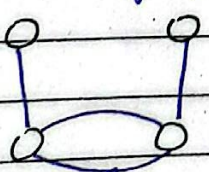
c) Not bipartite

d) Not bipartite

21. G: Yes, can be divided into: $\{u, w\}$ & $\{v, t, x\}$
H: Yes, can be divided into: $\{a, d, c\}$ & $\{b, e, f\}$
A: No, has an odd cycle
B: Yes, $\{d\}$, $\{e, f, g, h, i, j, k, l\}$

22. i) 

23. a) No such graph exists as sum of degrees is odd number

b) 

c) No such graph exists. Vertices of 3 connect to everyone else, forcing all degrees to be 2.

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24 a) No, number of vertices with odd degree must be even.

b) Yes, as the number of vertices with odd degree is even

c) $\Sigma \deg(v) = 10 \times 9 = 90$ Σ edges $= 90/2 = 45$
This scenario model K_{10} (complete graph).

d) Possible to be friends with 2 people as the sum of degree will be even.
Being friends with 3 people not possible as sum of degree ~~of odd~~ will be odd.

e) $4V = 2(10)$ $\therefore V = 5$

25. a) W and Z connected
Y: {A, D, E} & {B, C}
X: {A, B, E} & {C, D}

b) W ~~and X~~ has not a cycle

c) W, Y and Z

d) W and Y

26. a) $w_1 = v_3$ $w_5 = v_4$ $w_2 = v_1$ $w_3 = v_2$ $w_4 = v_5$

b) $u_1 = v_5$ $u_5 = v_1$ $u_4 = v_3$ $u_2 = v_2$ $u_3 = v_4$ $u_6 = v_6$

c) ~~Not isomorphic~~
~~as in 6~~
$u_1 = v_6$ $u_6 = v_7$ $u_7 = v_5$ $u_4 = v_2$ $u_3 = v_3$ $u_2 = v_4$ $u_5 = v_1$

27. a)

Path	d(b)	d(c)	d(d)	d(e)	d(f)	d(z)
a	12,a	(10,a)	-	-	12,a	
ac	(12,a)	21,c	13,c	(2,a)	-	
acf	(12,a)	21,c	13,c		21,f	
acfb		20,c	(13,c)		21,f	
acfbe	-	-	21,c		(19,e)	

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b)

Path	d(b)	d(c)	d(d)	d(e)	d(f)	d(g)	d(z)
a	4,a	(3,a)	-	-	-	-	-
ab	(4,a)		6,c	9,c	-	-	-
acb		(6,c)	9,c	-	-	-	-
acbd		-	(7,d)	11,d	-	-	
acbde				(11,d)	12,e		
acbdef					(12,e)	18,f	
acbdefg						(16,g)	

28.

a) {A, B, C, D, A}

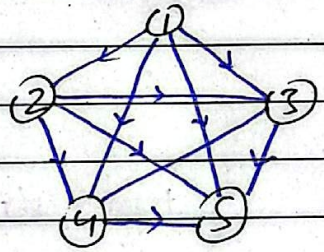
b) {A, B, C, D, A}

c) {A, B, D, C, A}

29. Yes, it is possible to find a trail

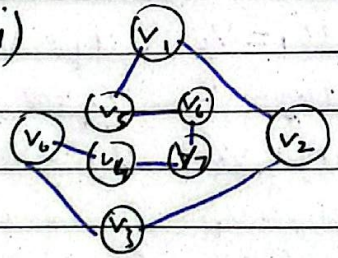
{A, G, H, F, E, H, I, E, K, J, D, C, B}

30

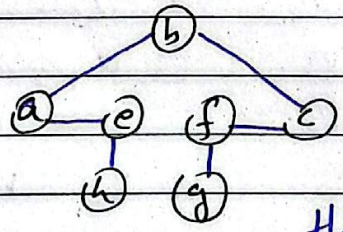


It is a eulerian circuit since every vertex has even degree

31. i)

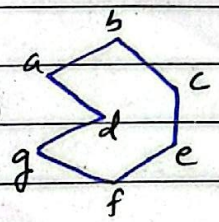


ii)



Hence, hamilton path and graph not possible

iii)



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32. i) has hamiltonian circuit : a, b, c, d, e, f, g, h, a

ii) ~~does not~~ have hamiltonian circuit : ~~j, h, a, b, c~~

iii) has hamiltonian circuit as it is a complete bipartite

33. i) $V_1, V_2, V_5, V_4, V_5, V_2, V_3, V_4, V_1$

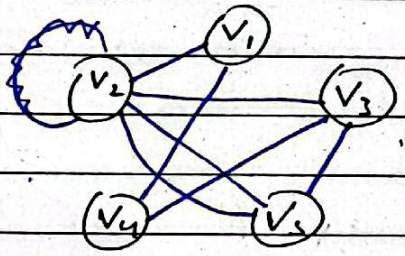
ii) Does not have a Eular circuit as V_4, V_7 & V_1 have odd degree

iii) Not a Euler circuit, e has odd degree

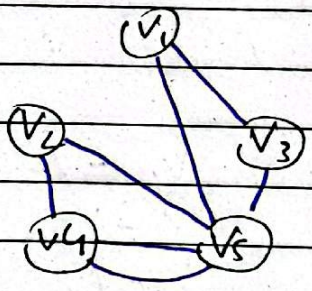
34. i) ~~Not, e~~ has 4 nodes with odd degrees so it is not Eulerian

ii) $u, V_7, V_0, V_1, u, V_2, V_4, V_3, V_2, V_6, V_5, w, V_4, V_6, w$

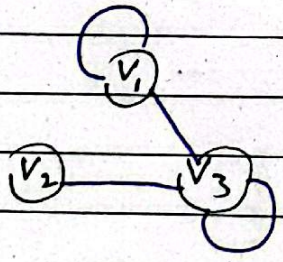
35. iii)



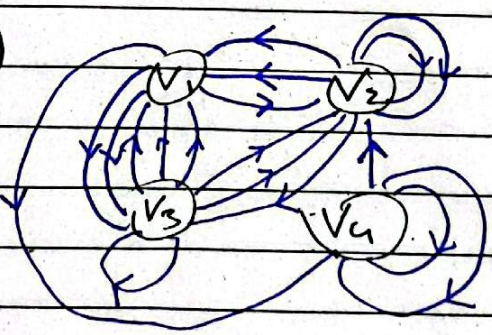
ii)



36. i)



ii)



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t not

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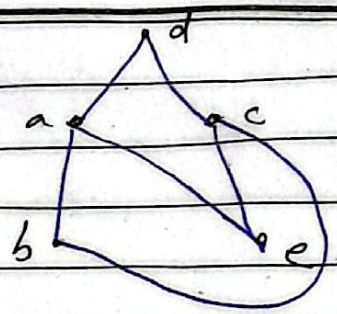
37. i) $e \leq 2v - 4$

$6 \leq 5 - 4$

$6 \leq 6$



ii) $e \leq 2v - 4$

$9 \leq 12 - 4$

$9 \not\leq 8$ It is not planar.

38.

	Vert.	Edges	Regions	Euler's
G	4	6	4	
H	6	9	5	
I	5	10	7	

39. a) 3 b) 1

c) 5 d) u, v

e) d f) k, l

g) m, s, t, x, y h) nodes with children

i) v, n, h, d, a j) vertices with no children

40. i) $v_0 - v_5$

$v_5 - v_6$

$v_6 - v_7$

$v_7 - v_4$

$v_4 - v_3$

$v_3 - v_1$

ii)